Mengmeng Wang

Data Scientist, Biomedical Engineer, Researcher

+61-466-989-626
mengmeng.wang.phd@gmail.com
/mengmwang
/mengmwang
/mengmwang.github.io

Profile

I am a dedicated data scientist, biostatistician & statistical signal processing researcher with a background in biomedical engineering and computer science. I have worked with health professionals and researchers at Orygen, Royal Children's Hospital and Bionic Institute on mental health and neuroscience projects. I have developed time-series analysis, predictive modelling, raw data transformation & analysis, signal processing, machine learning and image processing algorithms. I am experienced in Python, R, MATLAB, Excel, Power BI, Tableau, and various Python and R libraries.

Skills

Programming Languages Python, R, SQL, MATLAB

Technical Tools GitHub, Tableau, Power BI, Excel

Python Libraries Pandas, NumPy, Scikit-Learn, NLTK, SciPy, Matplotlib, JSON, RegEx

Professional Data Analysis, Technical Writing, Project Management, Teamwork, Communications

Education

Doctor of Philosophy – Statistical Signal Processing 2017 – 2023
The University of Melbourne

Master of Science – Image Processing 2013 – 2014
University of Bristol

Bachelor of Engineering – Telecommunications Engineering 2009 – 2013
Beijing University of Posts and Telecommunications

Experience

Research Fellow – Data Scientist Feb. 2023 – Present
Centre for Youth Mental Health, Orygen

- Experienced in data extraction, wrangling, and statistical analysis using different types of data.
- Applied data-driven techniques, modern natural language processing and machine learning algorithms in data analysis and health outcomes prediction.
- Contributed to developing statistical analysis plans and drafting reports and research papers.

Data Processing & Machine Learning Tutor (Casual) Mar. 2019 – Nov. 2022
The University of Melbourne

- Tutored 3 university-level subjects: elements of data processing, machine learning, signals and systems.
- Key topics: data wrangling, format, visualisation, natural language processing, supervised and unsupervised learning, classification, regression, clustering, neural network.
- Used Python and Pandas, NumPy, Scikit-Learn libraries to teach modern data science and advanced machine learning concepts.

Consulting Engineer May 2015 – Apr. 2017
China Academy of Information and Communications Technology

- Analysed domestic and international technologies in telecommunications and information science.
- Provided strategic advice on markets and created technical models and evaluation tools.
- Produced over 20 consulting reports and data analysis deliverables to governmental and industrial clients.

Selected Projects (Available on GitHub)

EEG Data Analysis in Music Therapy

- Performed EEG Data Analysis and visualisations in music therapy research.
- EEG Data importing, cleaning, preprocessing, feature engineering, visualisations and statistical analysis.
- Collaborated with health professionals, doctors and music therapists, to investigate the impacts of music therapy on brain state and functional brain connectivity.
- Tools used: MATLAB and EEGLAB Toolbox.

Financial Timeseries Processing and Forecast

- Performed financial data analysis in Level 1 Limit Order Book data, across data processing pipeline.
- Data cleaning and preparation: outlier detection, data visualisation and feature engineering.
- Data analysis: correlation, moving-average, auto-regression analysis.
- Timeseries forecasting: auto-regression model and machine learning models (decision tree, logistic regression, neural networks).
- Tools used: Python, Jupyter Notebook, Pandas, NumPy, matplotlib, seaborn, statsmodels, sklearn, keras.

Customer Purchasing Behaviours Analysis

- Analysed customer transaction and purchase behaviour data to find patterns in customer behaviours.
- Perform data cleaning, preparation and visualisation to facilitate analysis.
- Draw insights on change in store layout in customer behaviours.
- Tools used: Python, Jupyter Notebook, Pandas, NumPy.

Reinforcement Learning and Multi-Armed Bandits (MABs)

- Implementation and evaluation of state-of-art reinforcement learning algorithms: ϵ-Greedy, Upper Confidence Bound (UCB), LinUCB and KernalUCB.
- Tools used: Python, Jupyter Notebook, NumPy.

Publications

- **M. Wang**, C. Davey and L. Johnston, "Correction of induced functional connectivity in filtered resting state fNIRS data," *The 27th Annual Meeting of the Organization for Human Brain Mapping (OHBM)*, 2021.
- **M. Wang**, L. Johnston and C. Davey, "Correction for time-varying signal power in fNIRS connectivity analyses," *Society of fNIRS Virtual Conference*, 2021.
- **M. Wang** and A. Seghouane, "Motion Artefact Removal in Functional Near-infrared Spectroscopy Signals Based on Robust Estimation," *IEEE International Conference on Acoustics, Speech and Signal Processing*, 2019.
- **M. Wang**, F. Zhang and D. Agrafiotis, "A very low complexity reduced reference video quality metric based on spatio-temporal information selection," *IEEE International Conference on Image Processing*, 2015.

Certificates and Awards

- Online Course Certificates in Data Science 101, Data Science Tools, Big Data 101, Introduction to Cloud, Blockchain Essentials, R Programming Language
- HackerRank Programming Certificates for Python and SQL
- Virtual Internship Certificates from ANZ Australia, Goldman Sachs, JPMorgan Chase & Co.
- Virtual Internship Certificates from KPMG Australia, Deloitte Australia, Quantium, General Electric
- Conference Travel Scholarships, The University of Melbourne
- Melbourne Research Scholarships, The University of Melbourne
- Excellent Examination Performance Prize, University of Bristol

Volunteer Experience

Girl Power Mentor Mar. 2021 – Dec. 2022

The University of Melbourne

- Mentored Year 11/12 female students with interests in science and engineering.
- Coached girls to develop strength and confidence to pursue tertiary education in science and engineering.

References

Available Upon Request